

$$T(3) \quad p(x) = \begin{cases} e^{-x/\theta}/\theta, & x \geq 0 \\ 0, & x < 0 \end{cases} \quad \theta > 0 \quad n=3.$$

$$\bar{\theta}_1 = \bar{x}$$

$$\bar{\theta}_3 = X_{(2)} \text{ - пор. статистика}$$

$$\begin{aligned} \mu_{\xi} &= \int_{-\infty}^{\infty} x \cdot dF(x) = \int_0^{\infty} x \cdot \frac{e^{-x/\theta}}{\theta} dx = \frac{1}{\theta} \int_0^{\infty} x \cdot e^{-x/\theta} dx \\ &= \frac{1}{\theta} \cdot (-\theta) \int_0^{\infty} x \cdot d(e^{-x/\theta}) = -\frac{x \cdot e^{-x/\theta}}{\theta} \Big|_0^{\infty} + \int_0^{\infty} e^{-x/\theta} dx = -\theta. \\ \int_0^{\infty} d(e^{-x/\theta}) &= \underline{-\theta} \end{aligned}$$

$$\begin{aligned} \mu_{\xi^2} &= \int_0^{\infty} x^2 \frac{e^{-x/\theta}}{\theta} dx = \frac{1}{\theta} \int_0^{\infty} x^2 d(e^{-x/\theta}) = - \int_0^{\infty} x^2 d(e^{-x/\theta}) = \\ &= 2 \int_0^{\infty} e^{-x/\theta} x dx = \underline{2\theta^2}. \end{aligned}$$

$$\underline{D_{\xi}} = \mu_{\xi^2} - (\mu_{\xi})^2 = 2\theta^2 - \theta^2 = \underline{\theta^2}$$

a) несмещу: $\forall \theta \in \Theta \quad \mu \bar{\theta} = \theta.$

$$\bar{\theta}_1: \mu \bar{\theta}_1 = \mu \left(\frac{1}{n} \sum x_i \right) = \frac{1}{n} \cdot n \mu_{\xi} = \underline{\theta} \text{ - несмещу}$$

$$\bar{\theta}_3: \mu \bar{\theta}_3 = \mu(X_{(2)})$$

$$\mu(X_{(2)}) = \int_0^{\infty} x \cdot p_2(x) dx$$

$$p_2(x) = \frac{n!}{(n-2)!} F(x) \cdot (1-F(x))^{n-2} \cdot p(x).$$

$$\begin{aligned} F(x) &= \int_0^x p(t) dt = \frac{1}{\theta} \int_0^x e^{-t/\theta} dt = -1 \cdot e^{-t/\theta} \Big|_0^x = -1 \cdot (e^{-x/\theta} - 1) = \\ &= \underline{1 - e^{-x/\theta}} \end{aligned}$$

$$p_2(x) = n(n-1) \cdot (1 - e^{-x/\theta}) \cdot (e^{-x/\theta})^{n-2} \cdot \frac{e^{-x/\theta}}{\theta}$$

$$\begin{aligned} \mu(X_{(2)}) &= \int_0^{\infty} x \cdot (n(n-1)) \cdot \frac{1}{\theta} \cdot (1 - e^{-x/\theta}) \cdot (e^{-x/\theta})^{n-2} dx = \\ &= n(n-1) \cdot \frac{1}{\theta} \left(\int_0^{\infty} x \cdot (e^{-x/\theta}) dx - \int_0^{\infty} x \cdot (e^{-x/\theta})^n dx \right) = \end{aligned}$$

$$\int_0^\infty x \cdot e^{-\frac{x}{\theta}} dx = -\frac{\theta}{n-1} \int_0^\infty x \cdot d(e^{-\frac{x}{\theta}}) = -\frac{\theta}{n-1} \left(x \cdot e^{-\frac{x}{\theta}} \Big|_0^\infty - \int_0^\infty e^{-\frac{x}{\theta}} dx \right) = \frac{\theta}{n-1} \left(-\frac{\theta}{n-1} \int_0^\infty d(e^{-\frac{x}{\theta}}) \right) = -\left(\frac{\theta}{n-1} \right)^2 \cdot e^{-\frac{x}{\theta}} \Big|_0^\infty = \frac{(\theta)^2}{(n-1)}$$

$$\int_0^\infty x \cdot (e^{-\frac{x^n}{\theta}}) dx = -\frac{\theta}{n} \int_0^\infty x \cdot d(e^{-\frac{x^n}{\theta}}) = -\frac{\theta}{n} \left(x \cdot e^{-\frac{x^n}{\theta}} \Big|_0^\infty - \int_0^\infty e^{-\frac{x^n}{\theta}} dx \right) = -\frac{\theta^2}{n^2} \cdot e^{-\frac{x^n}{\theta}} \Big|_0^\infty = \frac{(\theta)^2}{n}$$

$$\Theta \quad \mu(X_{(2)}) = n/(n-1) \cdot \frac{1}{\theta} \left(\frac{(\theta)^2}{(n-1)} - \frac{(\theta)^2}{n} \right) = \frac{\theta^2 (n^2 - n^2 + 2n - 1)}{n/(n-1)} \cdot \frac{1}{\theta} = \theta \frac{2n-1}{n(n-1)} = \text{answ.}$$

$$\tilde{\Theta}_3' = \frac{n(n-1)}{2n-1} X_{(2)}$$

$$\mu \tilde{\Theta}_3' = \frac{n(n-1)}{2n-1} \mu X_{(2)} = \theta.$$

$$r) \quad \mathcal{D} \tilde{\Theta}_1 = \mathcal{D} \left(\frac{1}{n} \sum X_i \right) = \frac{1}{n^2} \cdot n \cdot \mathcal{D} \xi = \frac{\theta^2}{n}$$

$$\mathcal{D} \tilde{\Theta}_3' = \mathcal{D} \left(\frac{n(n-1)}{2n-1} X_{(2)} \right) = \frac{n^2(n-1)^2}{(2n-1)^2} \mathcal{D} X_{(2)}$$

$$\mu X_{(2)}^2 = n/(n-1) \cdot \frac{1}{\theta} \left(\int_0^\infty x^2 \cdot (e^{-\frac{x}{\theta}}) dx - \int_0^\infty x^2 \cdot (e^{-\frac{x}{\theta}})^n dx \right) = \frac{n(n-1)}{\theta} \left(\int_0^\infty x^2 \cdot e^{-\frac{x}{\theta}} dx - \int_0^\infty x^2 \cdot e^{-\frac{x}{\theta}} dx \right) = -\frac{\theta}{n-1} \int_0^\infty x^2 \cdot d(e^{-\frac{x}{\theta}}) = -\frac{\theta}{n-1} \left(x^2 \cdot e^{-\frac{x}{\theta}} \Big|_0^\infty - 2 \int_0^\infty e^{-\frac{x}{\theta}} x dx \right) = \frac{2\theta}{n-1} \int_0^\infty e^{-\frac{x}{\theta}} x dx = \frac{2\theta^3}{(n-1)^2}$$

$$\int_0^\infty x^2 \cdot e^{-\frac{x^n}{\theta}} dx = \dots = \frac{2\theta}{n} \int_0^\infty x e^{-\frac{x^n}{\theta}} dx = \frac{2\theta^3}{n^3}$$

$$\mu X_{(2)}^2 = n(n-1) \cdot \frac{1}{\theta} \left(\frac{2\theta^3}{(n-1)^2} - \frac{2\theta^3}{n^3} \right) = \frac{2\theta^2 (n^3 - (n-1)^3)}{n^2(n-1)^2}$$

$$\mathcal{D} X_{(2)} = \frac{2\theta^2 (n^3 - (n-1)^3)}{n^2(n-1)^2} = \frac{\theta^2 (2n-1)^2}{n^2(n-1)^2} = \frac{\theta^2 (2n^2 - 2n + 1)}{n^2(n-1)^2}$$

$$= \frac{2\theta^2 (3n^2 - 3n + 1) - \theta^2 (4n^2 - 4n + 1)}{n^2(n-1)^2} = \frac{\theta^2 (6n^2 - 6n + 2 - 4n^2 + 4n - 1)}{n^2(n-1)^2} = \frac{\theta^2 (2n^2 - 2n + 1)}{n^2(n-1)^2}$$

$$\mathcal{D} \tilde{\Theta}_3' = \frac{n^2(n-1)^2}{(2n-1)^2} \cdot \frac{\theta^2 (2n^2 - 2n + 1)}{n^2(n-1)^2} = \theta^2 \frac{(2n^2 - 2n + 1)}{(2n-1)^2}$$

$$\frac{1}{n} < \frac{2n^2 - 2n + 1}{(2n-1)^2}, \quad n > 1$$

$\Rightarrow \bar{\theta}_1$ эффект $\bar{\theta}_3$.

б) Проверка модели на регулярность.

$\partial p(x, \theta)$ - неар. дифф на Θ .

$$2) I(\theta) = M \left[\left(\frac{\partial \ln p(x, \theta)}{\partial \theta} \right)^2 \right] = \int_0^\infty \left(\frac{\partial \ln p(x, \theta)}{\partial \theta} \right)^2 p(x, \theta) dx \quad \text{с}$$

$$\ln p(x, \theta) = \ln \left(\frac{e^{-x/\theta}}{\theta} \right) = -\frac{x}{\theta} - \ln \theta.$$

$$\frac{\partial \ln p(x, \theta)}{\partial \theta} = \frac{x}{\theta^2} - \frac{1}{\theta} = \frac{x - \theta}{\theta^2}.$$

$$\text{с} \int_0^\infty \left(\frac{x - \theta}{\theta^2} \right)^2 \cdot \frac{e^{-x/\theta}}{\theta} dx = \frac{1}{\theta^5} \int_0^\infty (x - \theta)^2 e^{-x/\theta} dx.$$

$$\begin{aligned} \int_0^\infty (x - \theta)^2 e^{-x/\theta} dx &= -\theta \int_0^\infty (x - \theta)^2 d(e^{-x/\theta}) = -\theta \left((x - \theta)^2 e^{-x/\theta} \Big|_0^\infty - \right. \\ &\quad \left. - 2 \int_0^\infty e^{-x/\theta} (x - \theta) dx \right) = \theta^3 + 2\theta(-\theta) \int_0^\infty (x - \theta) d(e^{-x/\theta}) = \\ &= \theta^3 - 2\theta^2 \left((x - \theta) e^{-x/\theta} \Big|_0^\infty - \int_0^\infty e^{-x/\theta} dx \right) = \theta^3 - \underline{2\theta^3 + 2\theta^3} \\ &= \underline{\theta^3} \end{aligned}$$

$$\Rightarrow I(\theta) = \frac{1}{\theta^5} \cdot \theta^3 = \frac{1}{\theta^2} > 0, \quad \text{неар.}$$

$$3) \int_{\Theta(\theta)} \frac{\partial p(x, \theta)}{\partial \theta} dx = 0 \quad \frac{\partial p(x, \theta)}{\partial \theta} = \frac{x}{\theta^3} e^{-x/\theta} - \frac{1}{\theta^2} e^{-x/\theta}$$

$$\int_0^\infty e^{-x/\theta} \left(\frac{x - \theta}{\theta^3} \right) dx = \frac{1}{\theta^3} \int_0^\infty e^{-x/\theta} (x - \theta) dx = 0$$

$\Rightarrow$ модель регулярна.

Проверка оценок на регулярность:

$$1) \bar{\theta}_1 = \bar{x}.$$

несмещу

$$D(\bar{\theta}_1) - \text{оцр на } \forall \text{ комбинации } - \text{до } \frac{\theta^2}{n} \text{ оцр.}$$

$\Rightarrow$ т.к. дост. усл. регул. оценки выполняются, можно применить кр. - Rao

$$I(\theta) \geq \frac{1}{n \cdot I(\theta)}$$

$$\frac{\theta^2}{n} = \frac{1}{n \cdot \frac{1}{\theta^2}} \Rightarrow \text{по дост. усл. оценка эффективна}$$

$$2) \bar{\theta}_3 = X_{(2)}$$

- иначе, условие регул. оценки не работает.

$$\bar{\theta}_3' = \frac{n(n-1)}{2n-1} X_{(2)}$$

$$I(\bar{\theta}_3') = \theta^2 \left(\frac{2n^2 - 2n + 1}{(2n-1)^2} \right) \quad - \text{огр. на 4 коэффициента}$$

примен. кр-во Крамера - Рао

$$\theta^2 \left(\frac{2n^2 - 2n + 1}{(2n-1)^2} \right) > \frac{1}{n \cdot \frac{1}{\theta^2}}$$

- ~~неэффективна~~

ниже не след,
дост. усл не работ.

Т4

равн. распр. к-з (-1, 1)

$$X \sim R(-1, 1) : p(x) = \frac{1}{2} \chi_{(-1, 1)}$$

дискр. в 1. (0) и (2).

Если θ - вер-ть попадания в непр. часть, то $1-\theta$ в дискр. часть

$$X \sim p(x, \theta) = \frac{\theta}{2} \chi_{(-1, 1)} + \frac{1-\theta}{2} \{0\} + \frac{1-\theta}{2} \{2\}$$

$\theta \in (0, 1)$

- вер. модель

$$M\xi = \int_{-\infty}^{\infty} x dF = \int_{-1}^1 \frac{\theta}{2} x dx + 0 + 2 \left(\frac{1-\theta}{2} \right) =$$

$$= \frac{1-\theta}{2}$$

$$M\xi^2 = \int_{-\infty}^{\infty} x^2 dF = \int_{-1}^1 \frac{\theta}{2} x^2 dx + \frac{1}{4} \left(\frac{1-\theta}{2} \right) = \frac{\theta}{3} + 2 - 2\theta = 2 - \frac{\theta}{3}$$

$$D\xi = M\xi^2 - (M\xi)^2 = 2 - \frac{\theta}{3} - \left(\frac{1-\theta}{2} \right)^2 = 2 - \frac{\theta}{3} - 1 + 2\theta - \frac{\theta^2}{4} =$$

$$= \frac{-\theta^2}{4} + \frac{\theta}{3} + 1$$

а) μ -г моментов (о.м.м.)

$$L_1(\theta) = L_1$$

$$1-\theta = \bar{x} \Rightarrow \bar{\theta}_1 = 1-\bar{x}$$

Максим. правдоподобие (ОМП).

$\bar{x}_n$ - выборка,

m - числа в 0 и 1

$n-m$ - из $(-1, 1)$

$$L(\theta) = \left(\frac{\theta}{2}\right)^{n-m} \left(\frac{1-\theta}{2}\right)^m$$

$$\ln(L) = (n-m) \ln \theta + m \ln \left(\frac{1-\theta}{2}\right)$$

$$(\ln L)' = \frac{n-m}{\theta} + \frac{m(-\frac{1}{2})}{\frac{1-\theta}{2}} = \frac{n-m}{\theta} - \frac{m}{1-\theta}$$

$$\frac{n-m}{\theta} - \frac{m}{1-\theta} = 0$$

$$\frac{(1-\theta)(n-m)}{\theta(1-\theta)} = 0$$

$$n-m - \theta n + \theta m - \theta m = 0 \Rightarrow \hat{\theta}_1 = 1 - \frac{m}{n} = 1 - \bar{x}$$

$$(\ln L)'' = -\frac{n-m}{\theta^2} - \frac{m}{(1-\theta)^2} < 0 \text{ - макс}$$

а) $\hat{\theta}_1 = 1 - \bar{x}$

Несм: $M\hat{\theta}_1 = M(1 - \bar{x}) = 1 - M\bar{x} = 1 - \frac{1}{n} M \sum x_i =$
 $= 1 - \frac{1}{n} \cdot n \cdot M\theta = 1 - 1 + \theta = \theta$ - несмещ

Сог: $D\hat{\theta}_1 = D(1 - \bar{x}) = D\bar{x} = D\left(\frac{1}{n} \sum x_i\right) = \frac{1}{n^2} \cdot n D\theta =$
 $= \frac{-\theta^2 + \theta + 1}{3n} \xrightarrow{n \rightarrow \infty} 0$ - сог по дост. услов

$$\hat{\theta}_2 = 1 - \frac{m}{n}$$

$$M\hat{\theta}_2 = \theta$$

$$M\hat{\theta}_2 = 1 - \theta$$

$$D\hat{\theta}_2 = \frac{\theta(1-\theta)}{n}$$

$$D\hat{\theta}_2 = \frac{(1-\theta)\theta}{n}$$

Несмещ-ть: $M\hat{\theta}_2 = M\left(1 - \frac{m}{n}\right) = M(1 - \bar{x}) = 1 - M\bar{x} = 1 - \frac{1}{n} \cdot n \cdot \theta = 1 - (1 - \theta) = \theta$ - несмещ

Сог: $D\hat{\theta}_2 = D(1 - \bar{x}) = D\bar{x} = \frac{(1-\theta)\theta}{n} \xrightarrow{n \rightarrow \infty} 0$

сог по дост. усл

б). Проверка на регулярность.

$$p(x, \theta) = \frac{\theta}{2}, \quad p_k(\theta) = \frac{1-\theta}{2}$$

дифф по θ на $(0, 1)$

$$\int_{A(\theta)} \frac{\partial p}{\partial \theta} dx + \sum_k \frac{\partial p_k}{\partial \theta} \geq 0$$

$$\frac{1}{2} \int_{-1}^1 dx + \left(-\frac{1}{2}\right) \cdot 2 = \frac{1}{2} \cdot 2 - 1 = 0$$

$$I(\theta) = M \left[\left(\frac{\partial \ln p(x, \theta)}{\partial \theta} \right)^2 \right] = \int_{A(\theta)} \left(\frac{\partial \ln p(x, \theta)}{\partial \theta} \right)^2 p(x, \theta) dx + \sum_k \left(\frac{\partial \ln p_k(\theta)}{\partial \theta} \right)^2 p_k(\theta)$$

$$= \int_{-1}^1 \frac{1}{\theta^2} \cdot \frac{\theta}{2} dx + \frac{1}{(1-\theta)^2} \cdot \frac{1-\theta}{2} + \frac{1}{(1-\theta)^2} \cdot \frac{1-\theta}{2} = \frac{1}{\theta} + \frac{1}{1-\theta}$$

$$= \frac{1}{\theta} + \frac{1}{1-\theta} = \frac{1}{\theta(1-\theta)} > 0 \text{ некр}$$

$\Rightarrow$ модель регул

$$\bar{\theta}_1 = 1 - \bar{x}$$

по дост. услов. несмещ. модель регул.

$$D(\bar{\theta}_1) = \frac{\theta^2 + \frac{\theta}{n}}{n} - \text{огр на } \theta \text{ из } (0,1)$$

$\Rightarrow$ оценка регулярн.

некр во кр - Rao

$$D(\bar{\theta}_1) \geq \frac{1}{n \cdot I(\theta)}$$

$$\frac{\theta^2 + \frac{\theta}{n}}{n} \geq \frac{1}{n \cdot \frac{1}{\theta(1-\theta)}}$$

~~не эффектив~~
никого не след, дост. усл не работает

$$\bar{\theta}_2 = 1 - \frac{n}{n}$$

оценка регул (по дост. услов.

$\frac{(1-\theta)\theta}{n}$ - огр на θ из $(0,1)$

кр - Rao

$$\frac{(1-\theta)\theta}{n} \geq \frac{1}{n \cdot \frac{1}{\theta(1-\theta)}}$$

$\Rightarrow$

оценка эффективна по дост. условию